EX NO:	Chebyshev IIR Filter
DATE:	2.1. Design and Performance Evaluation of a Chebyshev Type-I IIR Low-Pass Filter using MATLAB

clc; clear;

Wp=0.4; Ws=0.55; Rp=1; As=40;

EX NO:2

Chebyshev IIR Filter

DATE: 2.1. Design and Performance Evaluation of a Chebyshev Type-I IIR Low-Pass

Filter using MATLAB

[N,Wn]=cheb1ord(Wp,Ws,Rp,As);

[b,a]=cheby1(N,Rp,Wn);

fprintf('Filter Order = %d\n',N);

fprintf('Cutoff Frequency = %.2f * pi rad/sample\n',Wn);

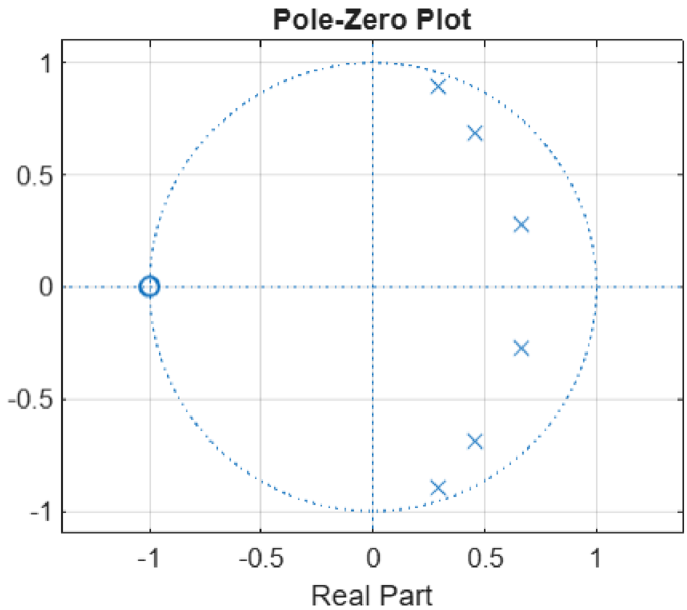
if all(abs(roots(a))<1)

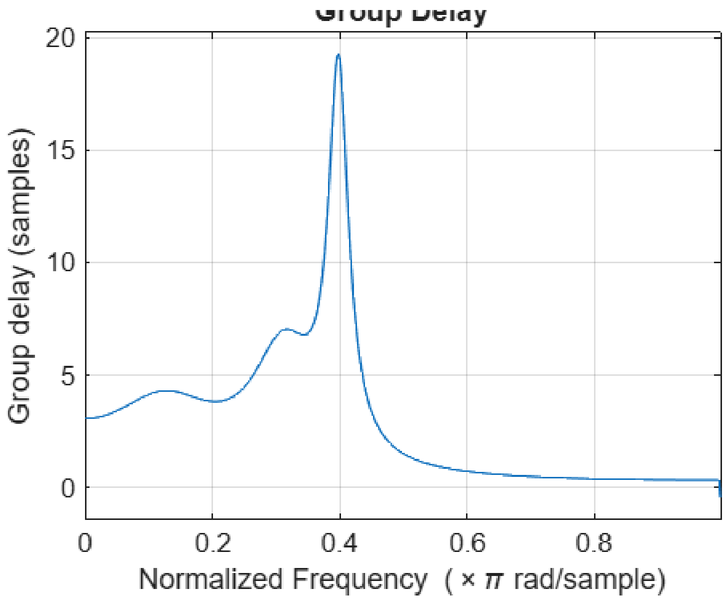
disp('System is STABLE');

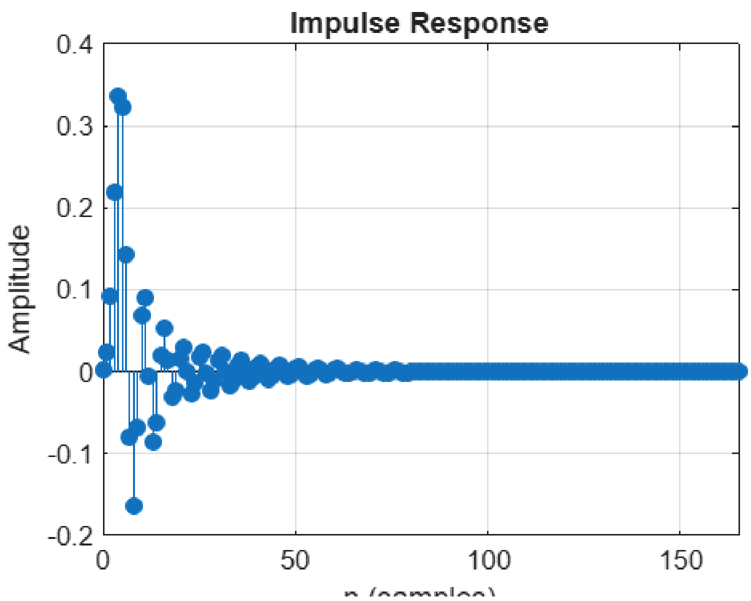
else

disp('System is UNSTABLE');

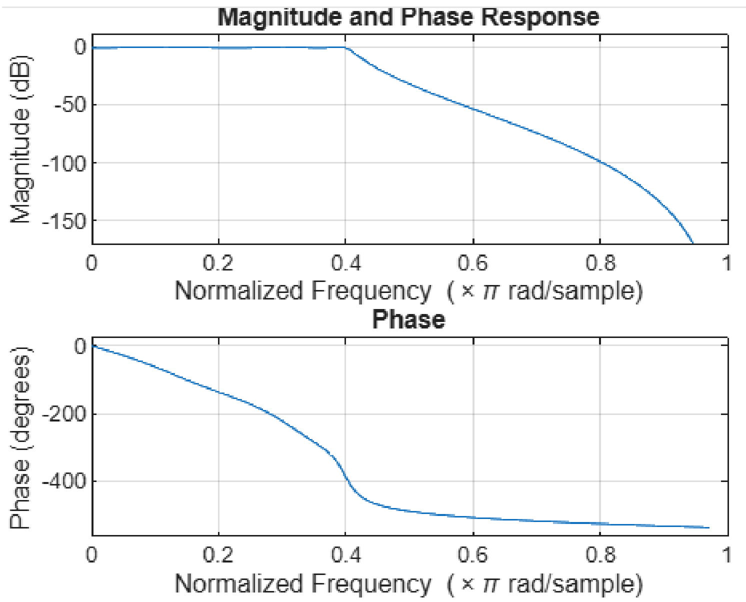
end


Pole-Zero Plot
1
0.5
0
-0.5
-1
-1
-0.5
0
0.5
1
Real Part


Group Delay
20
15
10
5
0
Group delay (samples)
0
0.2
0.4
0.6
0.8
Normalized Frequency (× π rad/sample)



2.mlx



Ex no 2	Chebyshev IIR Filter
date	2.2. Design and Performance Evaluation of a Chebyshev Type-I IIR High-Pass Filter

clc;

clear;

close all;

% Specifications

Rp = 1; % Passband Ripple (dB)

Rs = 40; % Stopband Attenuation (dB)

Wp = 0.5; % Passband Edge

EX NO:2 Chebyshev IIR Filter

DATE: 2.2. Design and Performance Evaluation of a Chebyshev Type-I IIR High-Pass

Filter

Ws = 0.35; % Stopband Edge

% Find Filter Order

[n,Wn] = cheb1ord(Wp,Ws,Rp,Rs);

% Design High-Pass Filter

[b,a] = cheby1(n,Rp,Wn,'high');

% Frequency Response

figure;

freqz(b,a);

title('Frequency Response');

% Impulse Response

figure;

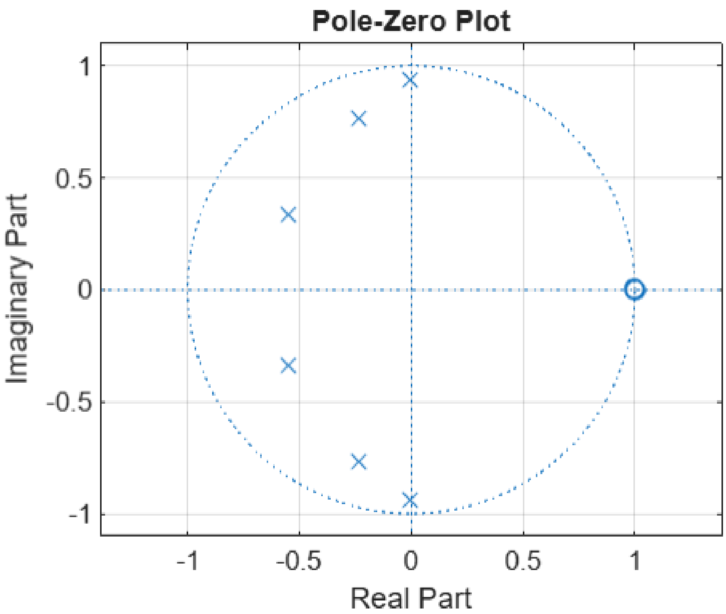
impz(b,a);

title('Impulse Response');

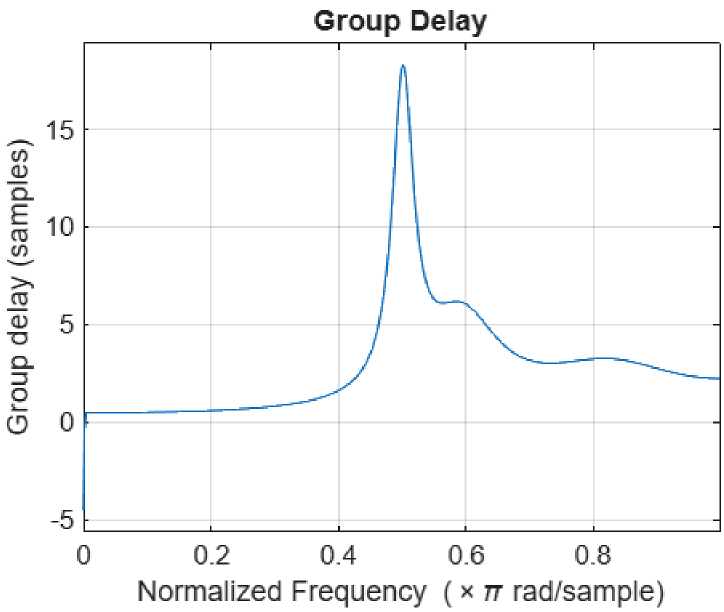
% Group Delay

figure;

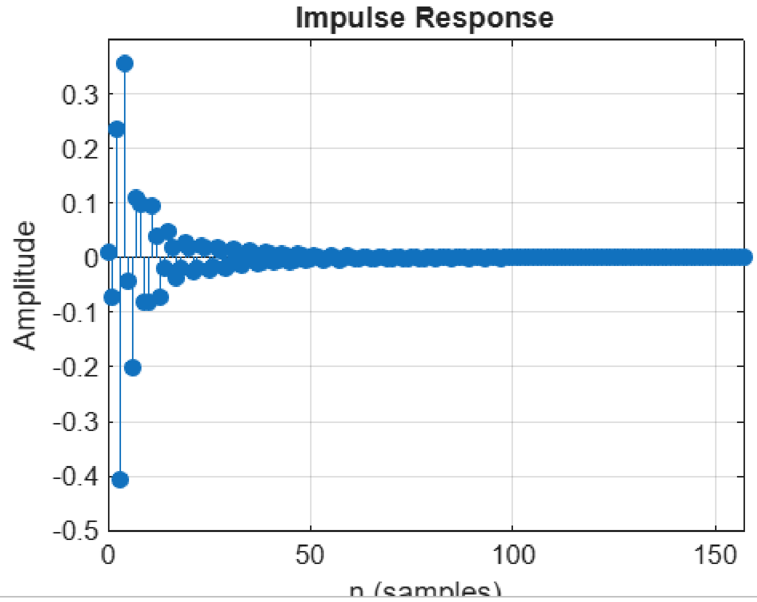
```
grpdelay(b,a);
title('Group Delay');

% Pole-Zero Plot
figure;
zplane(b,a);
title('Pole-Zero Plot');

fprintf('Filter Order = %d\n',n);

fprintf('Cutoff Frequency = %.4f\n',Wn);
```

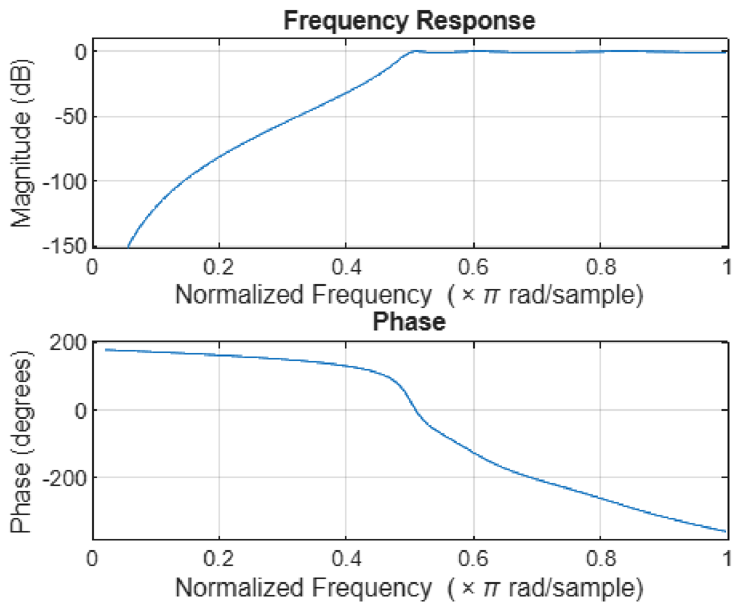


Pole-Zero Plot
Imaginary Part
Real Part
1
0.5
0
-0.5
-1
-1
-0.5
0
0.5
1



Group Delay
Group delay (samples)
Normalized Frequency (× π rad/sample)
15
10
5
0
-5
0
0.2
0.4
0.6
0.8


Impulse Response
Amplitude
0.3
0.2
0.1
0
-0.1
-0.2
-0.3
-0.4
-0.5
0
50
100
150
n (samples)


Frequency Response
Magnitude (dB)
0
-50
-100
-150
0
0.2
0.4
0.6
0.8
1
Normalized Frequency (× π rad/sample)
Phase
Phase (degrees)
200
0
-200
0
0.2
0.4
0.6
0.8
1
Normalized Frequency (× π rad/sample)